

Executive Summary & 3D ULPIN Encoding Standard:

Extends India's 14-digit Bhu-Aadhaar 2D cadastre into a robust 3D volumetric indexing standard for multi-storey apartments, subsurface metro/utility tunnels, and regulated air-rights corridors.

3D-ULPIN = IN-[14-Digit Base Parcel]-[Strata Zone]-[Floor Level]-[Volumetric Unit ID]-[Modulo97 Checksum]

Example: IN-2187-4930-1049-A-F12-U1204-K8 (Floor 12 • Apartment #1204)

1. RECOMMENDED PRODUCTION TECHNOLOGY STACK

3D WebGIS & Frontend

Next.js 14 / React 18 + TypeScript + Tailwind CSS

CesiumJS (Georeferenced 3D globe) + Three.js (Interior LoD-3 unit mesh, exploded view & strata slicing) + MapLibre GL.

Spatial Database & Topology

PostgreSQL 16 + PostGIS 3.4 + SFCGAL

Native 3D solid geometries ('POLYHEDRALSURFACEZ'), 'ST_3DIntersects', 'ST_Volume', 'ST_MakeSolid', and Redis tile caching.

AI/ML 3D Extraction Engine

Segment Anything (SAM) / YOLOv8-OBb (Footprints)

PDAL + Open3D + RANSAC 3D Plane Fitting (LiDAR floor height estimation) + IfcOpenShell (BIM IFC 4.3 parsing).

Geodetic Reference & APIs

FastAPI (Python 3.11) + GeoServer (OGC 3D Tiles)

EPSG:7755 (Indian National Grid) + WGS84 Orthometric Elevation linked to Survey of India CORS RTK Network.

2. STEP-BY-STEP 16-WEEK IMPLEMENTATION ROADMAP

Phase & Time	Key Milestones	Core Technical Deliverables	Tools & Stack
Phase 1 (W1-3) Data Ingestion	Multi-source spatial normalization & georeferencing	- Standardize on EPSG:7755 CRS & SoI CORS RTK coordinates. - Ingest drone photogrammetry orthomosaics & DSM/DTM. - Classify LiDAR point clouds (LAS/LAZ) via PDAL filter pipelines.	PDAL, GDAL, GeoPandas, LASpy
Phase 2 (W4-6) AI 3D Extraction	Automated building & floor boundary segmentation	- Extract building footprints using YOLOv8-OBb / SAM models. - RANSAC 3D plane fitting on LiDAR clusters for floor heights (Z). - Parse BIM IFC architectural files ('IfcSpace') to 3D volumes.	PyTorch, SAM, Open3D, IfcOpenShell
Phase 3 (W7-9) 3D ULPIN Core	Volumetric database & standard ULPIN generator	- Implement ISO 19152 compliant 3D ULPIN generation logic. - Setup PostGIS 3D database ('POLYHEDRALSURFACEZ' schema). - Automate precise volume calculation in cubic meters (m³).	PostgreSQL, PostGIS 3D, SFCGAL, FastAPI

3D ULPIN Implementation Roadmap (Continued)

PS ID: 26011

Phases 4–6 • Topology Validation, WebGIS Visualization & Open Standards Delivery

PHASE & TIME	KEY MILESTONES	CORE TECHNICAL DELIVERABLES	TOOLS & STACK
Phase 4 (W10–11) 3D Topology	Legal clash detection & manifold verification	- Volumetric overlap detection (<code>'ST_3DIntersects'</code> & <code>ST_Volume > 0</code>). - Topological water-tight manifold closure verification (<code>'ST_IsClosed'</code>). - Identify unregistered interior sliver gaps & cantilever overhangs.	PostGIS 3D, CGAL, Python Shapely
Phase 5 (W12–14) 3D WebGIS App	Interactive 3D Cadastre explorer & inspector	- Interactive 3D viewport (Three.js/Cesium) with raycast inspection. - Exploded vertical strata view & dynamic height clipping sliders. - Synchronized 2D elevation cross-section diagram rendering.	Next.js, Three.js, CesiumJS, Tailwind
Phase 6 (W15–16) Certification	3D Bhu-Aadhaar cards & open standard exports	- Generate official 3D Bhu-Aadhaar Digital Property Cards with QR. - Export datasets to OGC CityGML 3.0, ISO 19152 LADM & 3D GeoJSON. - Security audit, tamper-evident title hashing & registry deployment.	CityGML 3.0, LADM XML, GeoServer

3. CADASTRAL STRATA CLASSIFICATION MATRIX

STRATA LAYER	ELEVATION BOUNDS	3D ULPIN CODE	OWNERSHIP MODEL	CADASTRAL LOD
Air Rights	+45.0m to +65.0m	...-A-AIR-ZONE-A0	Statutory Airspace Right	LoD-1 Spatial Envelope
Residential Strata	+12.0m to +42.0m	...-A-F12-U1204-K8	Individual Freehold Title	LoD-3 Interior Polyhedron
Commercial Podium	+1.0m to +8.0m	...-A-F01-COMM-C3	Commercial Strata Title	LoD-3 Polyhedron
Surface Ground	0.00m Datum	...-S-S00-GRND-S1	Master Parent Parcel	LoD-2 Ground Footprint
Basement Parking	-6.5m to -1.0m	...-U-B01-PARK-B9	Common Strata Asset	LoD-2 Subsurface Solid
Metro Rail Tunnel	-21.0m to -14.0m	...-U-TUN-DMRC-T7	Public Statutory Easement	LoD-3 Subsurface Corridor